NAFISA TASNEEM ESHA

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Career Objective

To pursue graduate study and research in Computer Science and Engineering, with a focus on machine learning, bioinformatics, medical AI, and computer vision, and contribute to reliable and interpretable computational methods.

Education

Daffodil International University Jan 2023–Present
B.Sc. in Computer Science and Engineering **CGPA: 3.63/4.00**
Dhaka, Bangladesh

Cantonment Public School and College, Mymensingh 2020
Higher Secondary Certificate (HSC) [Certificate] **GPA: 5.00/5.00**

Aftab Uddin School and College 2018
Secondary School Certificate (SSC) [Certificate] **GPA: 5.00/5.00**

Research Interests

Machine learning; deep learning; computer vision and medical image analysis; bioinformatics and computational genomics; explainable AI; healthcare AI.

Undergraduate Research Experience

Undergraduate Thesis: PromBERT-X Ongoing
Explainable genomic language modeling for promoter prediction.

- Integrated global DNABERT-2 context with a CAMP local-context pathway for mammalian promoter prediction.
- Established a controlled cross-paradigm benchmark with redundancy control, heterogeneous negatives, and composition diagnostics.
- Extended evaluation with matched-seed analysis, calibration, efficiency assessment, human-to-mouse zero-shot transfer, and model-oriented attribution.

Research Member — mARS Lab, DIU Dec 2025–Present
Contribute to the CHASM gallbladder ultrasound classification project; author of the related manuscript under review at *Scientific Reports*. [Lab]

Research Assistant — Center for Advanced Research Jul 2026–Present
Support mentorship and training activities for students and new researchers in research learning and publication-oriented work.

Teaching Assistant — Department of CSE, DIU Jan 2025–Present
Support course-related academic activities and student learning under Ms. Shayla Sharmin, Assistant Professor, Department of CSE.

Publications & Accepted Papers

- Uddaula, A., Hasan, M., Sarker, D., **Esha, N. T.**, Hamim, M. S. H., & Amin, S. (2026). An Ensemble Machine Learning Framework for Malicious PDF Detection Using Static and Structural Features. *Proceedings of IDAA 2025*, Atlantis Press, 906–918. **Published.** [DOI]
- Hasan, M., Meraj, M. H., Tajbid, M., Hamim, M. S. H., **Esha, N. T.**, & Sarker, D. Machine Learning Approaches for Early Prediction of Chronic Kidney Disease and Its Progression. IDAA 2025; Taylor & Francis book chapter. **Accepted; publication pending.** [Certificate]
- Meraj, M. H., Hasan, M., Tajbid, M., Hamim, M. S. H., & **Esha, N. T.** (2026). PneuXNet: An Interpretable CNN Framework for High-Accuracy Pneumonia Detection Using Multi-Scale and Adaptive Attention. *UCICS 2026*, Varendra University, Rajshahi. **Accepted.** [Certificate]

Selected Manuscripts Under Review / Submitted

- Hasan, M., **Esha, N. T.**, & Assaduzzaman, M. PromBERT-X: An Explainable Global-Local Genomic Language Model for Promoter Prediction with Cross-Paradigm Benchmarking and Human-to-Mouse Zero-Shot Validation. *BMC Bioinformatics*. **Submitted.**
- Meraj, M. H., Hasan, M., Tajbid, M., **Esha, N. T.**, Assaduzzaman, M., Fahad, N., & Liew, T. H. CHASM: Overcoming Spatial Disruption with Linear-Time Hilbert State-Space Modeling for Gallbladder Ultrasound Classification. *Scientific Reports*. **Under review.**
- Tajbid, M., Hasan, M., Meraj, M. H., & **Esha, N. T.** Predicting Health Insurance Costs Using Machine Learning: A Regression Analysis of Demographic and Behavioral Risk Factors. *Journal of Emerging Global Health*. **Under review.**
- **Esha, N. T.**, Hasan, M., & Assaduzzaman, M. Explainable STRACE-RF for Cross-Architecture IoT Malware Detection under Unseen Architecture/Environment Domains. *IEEE Internet of Things Journal (IoT-J)*. **Submitted.**

Selected Projects

Credit Card Fraud Detection with Imbalanced Learning [Code]
Built an imbalance-aware fraud pipeline using class weighting, SMOTE, validation-based threshold tuning, PR-AUC/ROC-AUC evaluation, and Streamlit batch inference.

NLP Spam & Fake News Classification [Code]
Developed a leakage-aware TF-IDF classifier using Logistic Regression, calibrated Linear SVM, and Complement Naive Bayes with held-out evaluation and Streamlit inference.

AI-Powered Customer Support Ticket Review System [Code]
Built a human-in-the-loop NLP retrieval prototype that ranks relevant historical support tickets using text similarity.

Notice to Action [Code]
Built a Streamlit OCR and information-extraction application for deadlines, required actions, documents, conditions, and application details from notices, PDFs, and images.

CampusLab [Code]
Developed a Django laboratory discovery and booking system with authentication, conflict checking, equipment listing, approval workflows, and staff-side request review.

Technical & Research Skills

Programming: Python; working knowledge of C++, Java, and SQL.
AI / Data: TensorFlow, scikit-learn, OpenCV, DeepChem, NumPy, pandas, Matplotlib.
Methods: Data preparation, model implementation and evaluation, image preprocessing, NLP/information extraction, benchmarking, interpretability.
Tools: Git/GitHub, LaTeX, VS Code, Google Colab, Jupyter Notebook.
Languages: Bengali (native); English (research, presentations, and academic writing).

Leadership & Academic Engagement

Organizing Secretary — DIU Research Society Sep 2026–Present
Coordinate research-focused initiatives, seminars, workshops, and training; conducted *Research Foundations 101 with DIURS: Understanding Literature, Discovering Research Gaps*. [Certificate]

Executive Member & Coordinator (Lecture Series) — DIU Research Society Jun 2025–Sep 2026
Coordinated 20+ research and academic development programmes; supported team operations and guided junior members. [Certificate] [Program]

Girl Scout Second Leader — Coordinated squad activities and participated in social welfare and community-service initiatives. [Certificate]

Campus Ambassador & Volunteer — Robo Tech Valley, 6th International Robo Tech Olympiad — Selected to serve as a volunteer at Daffodil International University, Nov 2026.

Honors & Awards

- **2nd Runner-Up — Poster Presentation Competition** — *Reliability-Aware Spatiotemporal Analysis of Staple-Food Price Stress in Bangladesh: Evidence from 110 Markets*. 1st Conference on Innovation in Food, Health & Nutrition for Sustainable Development, DIU, 2026; Top 10 finalist among 81 participants. [Certificate]
- **Best Research Proposal Presenter Award** — "Meet the Researchers" Research Proposal Development Program, DIU Research Society, 2024. [Certificate]
- **Outstanding Performance, Research Project Showcasing** — Patent Idea Competition, Faculty of Graduate Studies, Daffodil International University, Nov 2025. [Certificate]

Academic References

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